

Blockchain Assignment 1

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Explore the following 4 websites and answer the questions

- bitcoin.org
- [Blockchain.com](https://blockchain.com)
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure? Blockchain
3. Bitcoin uses which consensus mechanism? Proof-of-Work
4. The smallest unit of Bitcoin is called? Satoshi
5. Bitcoin is a **peer-to-peer** type of what system? (Electronic) cash / payment system

Wallets

6. A wallet stores the user's **Private** key.
7. The recovery phrase used in wallets is called a **Seed** phrase.
8. Hardware wallets store keys **offline** (online/offline).

9. Bitcoin addresses are derived from **Public** keys.
10. What is required to sign a transaction? **Private (key)** (public/private)

Security

11. Bitcoin prevents double-spending using **Blockchain** (Proof-of-Work consensus).
12. Bitcoin is secured using **asymmetric** cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin **Core**.
14. The process of verifying transactions is called **Mining**.
15. The Bitcoin network is described as **Decentralized** (centralized/decentralized).

Mining

16. Miners solve **Cryptographic (hash)** puzzles to add blocks.
17. Mining rewards a block with newly created **Bitcoins**.
18. The block time target for Bitcoin is **10** minutes.
19. Mining difficulty adjusts every **2016** blocks.
20. Miners compete to find a valid block **Hash** (hash/signature).

Blockchain Structure

21. Each block contains a **Merkle** root.
22. Blocks are linked using the previous block's **Hash**.
23. A Bitcoin transaction contains inputs and **Outputs**.
24. Unspent transaction outputs are called **UTXOs**.

25. The first-ever Bitcoin block is called the **Genesis** block.

Network

26. A full node stores the entire blockchain **Locally**.

27. Bitcoin uses which protocol for communication? **P2P** (TCP/P2P)

28. A light wallet uses **SPV** mode (SPV/full).

29. Nodes relay information across a **P2P** network.

30. The maximum Bitcoin supply is **21** million.

Task B1: Explore a Block

Data taken from block #**961,122** on blockchain.com/explorer:

Field	Value
Block height	961,122
Block hash	000000000000000000000000c645aa044b2b4c9ccdc288ebfdfedf9b117818a6be
Timestamp	August 5, 2026, 12:35:03 AM (UTC)
Number of transactions	4,629
Miner/pool name	Unknown (the payout address wasn't tagged to a known mining pool by Blockchain.com; many blocks show "Unknown" when the pool doesn't sign its coinbase message with an identifiable tag)

Task B2: Explore a Transaction

TXID

f4184fc596403b9d638783cf57adfe4c75c605f6356fbc91338530e9831e9e16:

- TXID: f4184fc596403b9d638783cf57adfe4c75c605f6356fbc91338530e9831e9e16
- Number of inputs: 1
- Number of outputs: 1
- Transaction fee: 0 BTC (fees weren't customary in Bitcoin's earliest days)
- Confirmations: 900,000+ (it was mined in January 2009, so at block height ~961,000 today it has essentially the maximum possible confirmations — open the live page for the exact current count)

Task B3: Explore an Address

The block above was mined to address **1BM1sAcrfV6d4zPKytzziu4McLQDsFC2Qc**. Balances and totals on this address change constantly as new payouts arrive, so open the address page live and record:

- Total BTC received
- Total BTC sent
- Current balance

Role of a full node: A full node downloads and independently verifies the *entire* blockchain and every new transaction/block against Bitcoin's consensus rules. It doesn't trust any other party — it re-checks signatures, UTXO validity, and block rules itself — and then relays valid transactions/blocks to other peers. Full nodes are what make Bitcoin trustless and decentralized, since anyone running one can independently confirm they're seeing the real chain.

Two developer resources on bitcoin.org:

1. The **Bitcoin Developer Guide/Reference** (developer.bitcoin.org) — documentation on the protocol, P2P network, transactions, and wallets.
2. **Bitcoin Core** source code and release downloads (via the Bitcoin Core section), which developers use to run nodes or build on the reference client.

Explore: <https://coinmarketcap.com/>

Task C1: Market Information

Metric	Value
Current price of BTC	\$64,114.87
24-hour % change	+0.58%
Market capitalization	≈\$1.29 trillion (\$1,286,525,332,023)
Circulating supply	20,065,943 BTC
Total/maximum supply	21,000,000 BTC (~95.6% already mined)

Task C2: Historical Data

- Price one year ago (Aug 2025): Bitcoin was trading much higher, roughly \$108,000–\$115,000 through August 2025, closing the month around \$108,237. It later peaked at an all-time high of \$126,198.07 on October 6, 2025, then declined sharply over the following months to today's ~\$64,000 level (a drop of roughly 49% from the ATH).
- Last 30 days (approx., as of Aug 5, 2026): Trading has been range-bound in the low-to-mid \$60,000s 24h range alone was \$63,397 – \$64,467, and the broader recent range has been roughly \$61,000 (low) to \$66,000 (high). For the exact daily high/low, open CoinMarketCap's "Historical Data" tab for Bitcoin and set the date range to the last 30 days, since exact daily closes shift with each session.

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin's price swings heavily because it's a relatively young, thinly regulated asset with a fixed and inelastic supply; demand shocks (news, regulation, institutional buying/selling, macro events) move the price sharply since there's no central bank or mechanism to smooth supply. It also trades 24/7 across fragmented global exchanges, has a comparatively small market relative to traditional assets, and its price is heavily influenced by sentiment, speculation, and leverage in derivatives markets, all of which amplify moves in both directions.

List two investment risks mentioned.

Regulatory risk: governments can restrict, tax, or ban crypto trading/mining (e.g., China's 2021 crackdown), which can cause sudden, sharp price drops.

Security/custody risk: losing your private keys, exchange hacks, or scams can result in permanent, unrecoverable loss of funds, since there's no bank or FDIC-style insurance backing individual holdings.

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave the clearest conceptual understanding; it explains the "why" behind wallets, keys, mining, and nodes in plain language aimed at newcomers, without assuming you're trying to trade. Blockchain.com's explorer was best for hands-on understanding, since seeing a real block/transaction/address makes abstract terms like "UTXO" or "Merkle root" concrete. CoinMarketCap was best for market context (price, supply, cap), and Investopedia was best for the financial/risk framing. Overall, Bitcoin.org + the Blockchain.com explorer together gave the most complete picture; one explains the theory, the other lets you verify it live on-chain.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain replaces a trusted middleman (like a bank) with math, transparency, and distributed agreement. Every transaction is broadcast to a network of independent nodes, each of which verifies it against the same public rules, checking that funds actually exist (via UTXOs) and that the sender's cryptographic signature is valid before relaying it further. Verified transactions are bundled into blocks, and miners compete to add the next block by solving a computationally expensive Proof-of-Work puzzle; because altering any past block would require redoing that work for every block after it (an astronomically expensive feat once the chain is long), the ledger becomes effectively tamper-proof. Since thousands of nodes worldwide hold an identical copy of this chain and reject any block that breaks the rules, no single party not a bank, government, or even a majority of miners acting alone can unilaterally rewrite history or spend someone else's coins. Trust is thus produced collectively, through open verification and economic incentives, rather than through reliance on any one institution.